2025 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM) | 979-8-3315-3342-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/AIM64088.2025.11175780

LLM-Augmented Multi-Fidelity Bayesian Optimization for Parameter Optimization in Human-Robot Collaborative Assembly

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***Abstract*— Human-robot collaborative assembly (HRCA) is critical for manufacturing tasks like aircraft cabin drilling, where human contextual adaptability complements robotic repeatability. Parameter optimization—a critical component of HRCA systems—faces significant challenges due to limited high-fidelity experimental data, primarily resulting from costly physical trials and dynamic human-robot interaction uncertainties. While large language models (LLMs) show promise as low-cost estimators for parameter-quality relationships, their unreliable physical reasoning and hallucination-prone outputs limit direct application in safety-critical scenarios. To address these challenges, this paper proposes an LLM-augmented multi-fidelity Bayesian optimization framework for parameter optimization. This approach leverages LLMs' strengths while mitigating the scarcity of real samples and minimizing potential hallucination issues. First, a GPT-based LLM serves as the low-fidelity model, generating initial parameter-quality predictions using tailored prompts that encode domain-specific physics. Then, a latent variable Gaussian process (LVGP) hierarchically links the LLM's predictions with high-fidelity simulations through a shared covariance structure, enabling residual-based uncertainty propagation. Following this framework, an adaptive acquisition function prioritizes parameter combinations that maximize cross-fidelity consensus. Simultaneously, deviations between LLM and high-fidelity outputs trigger residual-guided prompt engineering to refine the LLM's reasoning. Failed optimization trials are systematically accumulated to iteratively retrain the LVGP kernel and adjust the LLM's prompting strategy. A case study of parameter optimization for robotic arm drilling in aircraft cabin assembly will demonstrate the advantages of the proposed method.**

I. INTRODUCTION

Human-robot collaborative assembly (HRCA) has revolutionized modern manufacturing by synergizing human ingenuity with robotic precision [1]. In HRCA systems, humans contribute adaptive decision-making, anomaly resolution, and contextual awareness, while robots excel in repetitive, high-precision tasks such as drilling, fastening, and part positioning. This collaboration is indispensable in industries like aerospace, automotive, and medical device manufacturing, where tasks demand both flexibility and micro-level accuracy.

Despite its advantages, HRCA is inherently volatile due to the unpredictable interplay between human actions, robotic operations, and environmental factors. Unlike static automation, HRCA parameters (e.g., robotic speed, force thresholds) must adapt in real time to disturbances such as sudden design changes, tool wear, or shifts in material properties [2]. For example, a worker adjusting a robot's drilling path mid-task to avoid an unexpected composite fiber orientation alters the system's operating conditions, invalidating pre-programmed parameters. Simultaneously, human interventions—though essential for adaptability—introduce variability that complicates parameter optimization. Workers often lack prior data for novel scenarios, forcing them to rely on iterative trial-and-error. Such volatility underscores the need for adaptive optimization frameworks that reconcile human flexibility with robotic consistency.

Traditional parameter tuning methods, such as rule-based grid search, struggle to address HRCA's dynamic demands. Grid search operates by testing fixed parameter combinations [3], guided by empirical rules like "maximum drilling speed for aircraft cabin hole." While simple, this approach assumes static conditions and fails to adapt to real-time perturbations like tool degradation or material inconsistencies. Its computational cost also scales exponentially with parameter dimensions, rendering it impractical for multi-variable tasks (e.g., concurrent optimization of speed, feed rate, and coolant flow).

Bayesian optimization (BO) offers a data-driven alternative by iteratively refining parameter choices using probabilistic surrogate models [4]. Unlike grid search, BO balances exploration of uncertain regions and exploitation of known optima, theoretically accelerating convergence. However, two barriers hinder its adoption in HRCA: 1. High sample costs: Each physical experiment (e.g., drilling trials) is time-intensive, requiring dozens of iterations for reliable results. 2. Cold-start dependency: BO relies on initial data to bootstrap optimization, which is unavailable for novel materials or processes.

Large language models (LLMs) pretrained on domain-specific knowledge (e.g., material science manuals, machining guidelines) can mitigate these barriers by generating synthetic parameter-performance data [5]. For example, an LLM might infer optimal drilling parameters for an untested CFRP-aluminum stack by extrapolating from similar material properties [6]. However, LLM suggestions risk inaccuracies ("hallucinations") due to ambiguous inputs or incomplete training data.

To harmonize these capabilities, LLM-Augmented Multi-Fidelity BO is proposed, integrating LLMs with BO techniques through a hierarchical uncertainty quantification framework. The main contributions of this work can be listed as:

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1. A methodological approach that adapts LLMs as physics-informed surrogate models for engineering optimization without domain-specific fine-tuning, effectively addressing the data scarcity challenge in parameter optimization.

2. A novel mathematical framework that systematically connects LLM-derived predictions with high-fidelity simulations through latent variable structures, enabling robust uncertainty quantification across different fidelity levels.

3. An iterative prompt refinement mechanism that leverages discrepancies between model predictions and simulation outcomes to enhance the physical reasoning capabilities of the optimization process.

II. Related work

A. Parameter Optimization

Parameter optimization serves as the computational backbone for modern manufacturing systems, with methodological evolution progressing from heuristic approaches to probabilistic reasoning frameworks.

Traditional parameter space design often inadequately addresses the curse of dimensionality and variable interaction complexities. Grid search methodologies [7] employ fixed spatial partitions that presume uniform parameter sensitivity - an assumption violated in most manufacturing processes exhibiting nonlinear parameter couplings. While random search [8] alleviates dimensional rigidity through stochastic sampling, empirical studies demonstrate that most of sampled points in high-dimensional spaces reside in non-optimal regions [9]. Gradient-based optimization fails catastrophically in mixed-variable environments common to industrial applications, such as simultaneous optimization of continuous thermal parameters and discrete material selections in materials design [10]. Evolutionary strategies compound these issues through uncontrolled parameter space expansion, with genetic algorithms generating physically infeasible solutions [11].

BO revolutionizes parameter discovery through its dual surrogate-acquisition function architecture. Gaussian process (GP) surrogates [4] enable uncertainty-aware sampling, achieving faster convergence than manual tuning in neural hyperparameter optimization [12]. The SMAC algorithm [13] addresses GP's cubic time complexity through random forest surrogates). Modern innovations like deep kernel learning [14] and asynchronous parallelization [15] extend BO's applicability to high-throughput industrial scenarios.

BO's efficacy in manufacturing environments remains contingent on acquiring high-fidelity process data. Persistent challenges including sensor noise contamination and temporal process drift induce non-stationary data distributions that systematically degrade surrogate model accuracy and optimization reliability.

B. LLM Applications and Hallucination Risks

LLMs present transformative potential for industrial automation, yet require rigorous safeguards against hallucination-induced failures in safety-critical systems.

Modern LLMs transcend textual processing through instruction-aware architectures. Chain-of-thought augmented models [5] achieve human-competitive performance in technical domains like equipment maintenance scheduling. Vision-language models [16] enable cross-modal feature alignment, permitting direct translation of quality control images to process parameter adjustments (Deng et al., 2023). Recent breakthroughs demonstrate LLMs' capacity to generate BO priors from equipment logs in production line tuning (Chen et al., 2023). Autonomous robotic control frameworks (Wang et al., 2023) further showcase LLMs' potential to interpret natural language directives into executable parameter sets.

LLMs and vision-language models (VLMs) [16] are revolutionizing industrial automation through multimodal understanding and decision-making capabilities. LLMs excel at parsing unstructured textual data, enabling automated analysis of technical documentation [17] and real-time interpretation of equipment maintenance logs. Their few-shot learning capacity allows rapid adaptation to domain-specific vocabularies, such as generating specific solution in Precise Machining [18]. Besides, Retrieval-augmented generation [19] grounds outputs in verified knowledge bases, while self-consistency verification (Wang et al., 2023) enforces consensus across multiple reasoning paths. VLMs bridge visual and linguistic reasoning, achieving human-level accuracy in defect classification [20]. Industrial deployments demonstrate VLMs' ability to guide assembly line adjustments based on real-time visual feedback, reducing manual inspection workloads [21]. These models further enhance workforce training through interactive simulations that combine 3D visualizations with natural language instructions, accelerating operator proficiency in complex machinery operations.

LLM-driven parameter optimization introduces unique failure modes where undetected hallucinations may cascade into catastrophic process deviations. This underscores the imperative for embedded verification systems and human-in-the-loop safeguards within industrial LLM deployments.

III. Theoretical Background: Bayesian Optimization

BO is a sample-efficient sequential optimization framework particularly suited for expensive black-box functions. The core components consist of a probabilistic surrogate model and an acquisition function that guides the search process.

A. Gaussian Process Regression

The Gaussian Process (GP) serves as the fundamental surrogate model in BO, providing Bayesian posterior estimation over the objective function $f : \mathcal{X} \to \mathbb{R}$. A GP is fully specified by its mean function $m(\cdot)$ and covariance kernel $k(\cdot, \cdot)$:

$$f(\mathbf{x}) \sim \mathcal{GP}\left(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')\right) \tag{1}$$

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For observed data $\mathcal{D}_{1:t} = \{(\mathbf{x}_i, y_i)\}_{i=1}^t$ with $y_i = f(\mathbf{x}_i) + \epsilon_i$ where $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, the posterior predictive distribution at new point $\mathbf{x}_*$ follows:

$$p(f(\mathbf{x}_*)|\mathcal{D}_{1:t}, \mathbf{x}_*) = \mathcal{N}\left(\mu_t(\mathbf{x}_*), \sigma_t^2(\mathbf{x}_*)\right) \tag{2}$$

Posterior mean and variance are given by:

$$\mu_t(\mathbf{x}_*) = \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}_{1:t} \tag{3}$$

$$\sigma_t^2(\mathbf{x}_*) = k(\mathbf{x}_*, \mathbf{x}_*) - \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_* \tag{4}$$

where $\mathbf{K} \in \mathbb{R}^{t \times t}$ is the kernel matrix with entries $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$, $\mathbf{k}_* = [k(\mathbf{x}_*, \mathbf{x}_1), \ldots, k(\mathbf{x}_*, \mathbf{x}_t)]^\top$, and $\mathbf{y}_{1:t} = [y_1, \ldots, y_t]^\top$.

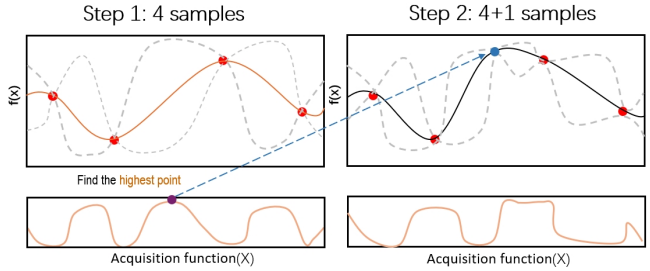


Fig. 1. BO principle explanation

B. Acquisition Functions

The BO workflow iterates between:

1) Surrogate modeling: Update GP posterior using all observations
2) Acquisition function maximization: Select next query point $\mathbf{x}_{t+1} = _{\mathbf{x} \in \mathcal{X}}\ \alpha_t(\mathbf{x})$

Common acquisition functions include:

Expected Improvement (EI):

$$\alpha_{\text{EI}}(\mathbf{x}) = \mathbb{E}\left[\max(f(\mathbf{x}) - f(\mathbf{x}^+), 0)\right] \tag{5}$$

where $\mathbf{x}^+ = _{\mathbf{x}_i \in \mathcal{D}_{1:t}}\ f(\mathbf{x}_i)$ is the incumbent solution.

Upper Confidence Bound (UCB):

$$\alpha_{\text{UCB}}(\mathbf{x}) = \mu_t(\mathbf{x}) + \kappa \sigma_t(\mathbf{x}) \tag{6}$$

with exploration parameter $\kappa > 0$ controlling the trade-off between exploration and exploitation.

IV. Methodology

A. Problem statement

Parameter optimization in HRCA faces significant challenges due to costly physical trials and uncertainties in human-robot interactions that severely limit high-fidelity experimental data, creating a substantial barrier to effective optimization. While LLMs offer potential as low-cost estimators for parameter-quality relationships, their direct application is constrained by unreliable physical reasoning and tendency toward hallucinations. A fundamental challenge lies in bridging the gap between these low-fidelity LLM outputs and sparse high-fidelity experimental data—a connection that requires addressing complex discrepancies in accuracy, reliability, and physical consistency between these fundamentally different information sources.

B. Collaboration of Multi-fidelity model

We propose a multi-fidelity Bayesian optimization (MFBO) framework that synergistically integrates high- and low-fidelity models through residual correction. The goal is to minimize costly high-fidelity evaluations while maintaining optimization accuracy. Key components include:

- Two BO models with *residual correction* to bridge fidelity gaps.
- An *adaptive candidate selection* rule based on expected improvement (EI).
- Iterative updates and convergence-driven sample size adaptation.

High-Fidelity Model (HFM): The HFM represents the true objective function $f_{\text{HF}}(\mathbf{x})$, where $\mathbf{x}$ is the parameter vector to optimize. Its surrogate is a Gaussian process (GP) trained on dataset $\mathcal{D}_{\text{HF}}$:

$$y_{\text{HF}}(\mathbf{x}) = f_{\text{HF}}(\mathbf{x}) + \epsilon_{\text{HF}}, \quad \epsilon_{\text{HF}} \sim \mathcal{N}(0, \sigma_{\text{HF}}^2), \tag{7}$$

where $y_{\text{HF}}(\mathbf{x})$ is the observed high-fidelity output, $f_{\text{HF}}(\mathbf{x})$ is the true high-fidelity function, ϵ_{HF} is additive Gaussian noise with variance σ_{HF}^2, and $\mathcal{D}_{\text{HF}} = \{(\mathbf{x}_i, y_{\text{HF},i})\}_{i=1}^{N_{\text{HF}}}$ is the historical HFM data.

Low-Fidelity Model (LFM): The LFM approximates f_{HF} with lower accuracy but faster evaluation. Its GP surrogate uses dataset $\mathcal{D}_{\text{LF}}$:

$$y_{\text{LF}}(\mathbf{x}) = f_{\text{LF}}(\mathbf{x}) + \epsilon_{\text{LF}}, \quad \epsilon_{\text{LF}} \sim \mathcal{N}(0, \sigma_{\text{LF}}^2), \tag{8}$$

where $y_{\text{LF}}(\mathbf{x})$ is the observed low-fidelity output, $f_{\text{LF}}(\mathbf{x})$ is the low-fidelity approximation of f_{HF}, ϵ_{LF} is additive noise with variance σ_{LF}^2, and $\mathcal{D}_{\text{LF}} = \{(\mathbf{x}_j, y_{\text{LF},j})\}_{j=1}^{N_{\text{LF}}}$ is the historical LFM data.

Residual Correction: To link the models, we define a residual term $r(\mathbf{x})$, modeled as a GP:

$$r(\mathbf{x}) = y_{\text{HF}}(\mathbf{x}) - y_{\text{LF}}(\mathbf{x}), \tag{9}$$

where $r(\mathbf{x})$ represents the discrepancy between HFM and LFM outputs. This residual corrects LFM predictions using HFM data, ensuring systematic errors are minimized.

The approach balances exploration-exploitation by selecting parameters that maximize expected improvement in either model, while effectively leveraging information shared across different fidelity levels.

1) **Candidate Selection:** Compute EI for both models at iteration t:

$$\text{EI}_{\text{HF}}(\mathbf{x}) = \mathbb{E}\left[\max\left(f_{\text{HF}}(\mathbf{x}) - f_{\text{HF}}^+, 0\right)\right], \tag{10}$$

$$\text{EI}_{\text{LF}}(\mathbf{x}) = \mathbb{E}\left[\max\left(f_{\text{LF}}(\mathbf{x}) - f_{\text{LF}}^+, 0\right)\right], \tag{11}$$

where $f_{\text{HF}}^+ = \max\{y_{\text{HF},i}\}$ and $f_{\text{LF}}^+ = \max\{y_{\text{LF},j}\}$ denote the best observed values of the HFM and LFM, respectively.

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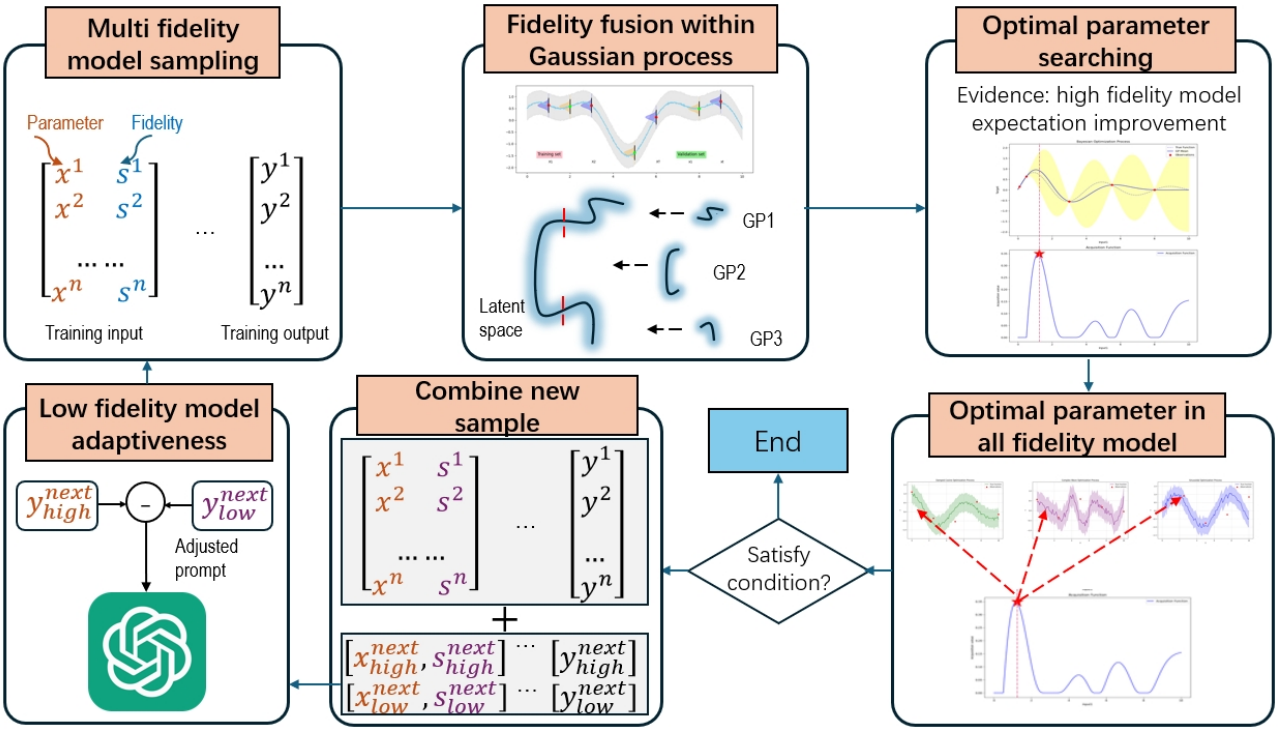


Fig. 2. Proposed method architechture

2) **Parameter Selection:** Choose the parameter with the largest EI:

$$\mathbf{x}_{t+1} = \arg\max_{\mathbf{x}} \left(\mathrm{EI}_{\mathrm{HF}}(\mathbf{x}), \mathrm{EI}_{\mathrm{LF}}(\mathbf{x})\right), \quad (12)$$

where $\mathbf{x}_{t+1}$ is the next candidate parameter to evaluate.

3) **Cross-Evaluation:** Evaluate $\mathbf{x}_{t+1}$ on both models:

$$y_{\mathrm{HF},t+1} = f_{\mathrm{HF}}(\mathbf{x}_{t+1}), \quad y_{\mathrm{LF},t+1} = f_{\mathrm{LF}}(\mathbf{x}_{t+1}), \quad (13)$$

obtaining new evaluations $y_{\mathrm{HF},t+1}$ and $y_{\mathrm{LF},t+1}$.

4) **Model Updates:** Augment datasets and retrain GPs:

$$\mathcal{D}_{\mathrm{HF}} \leftarrow \mathcal{D}_{\mathrm{HF}} \cup \{(\mathbf{x}_{t+1}, y_{\mathrm{HF},t+1})\}, \quad (14)$$

$$\mathcal{D}_{\mathrm{LF}} \leftarrow \mathcal{D}_{\mathrm{LF}} \cup \{(\mathbf{x}_{t+1}, y_{\mathrm{LF},t+1})\}. \quad (15)$$

5) **Convergence Check:** Terminate if:

$$\left| f^{+}_{\mathrm{HF}} - f^{+}_{\mathrm{HF},t} \right| < \tau \quad \text{or} \quad t \geq T_{\max}, \quad (16)$$

where τ is a user-defined convergence tolerance and $T_{\max}$ is the maximum allowed iterations. If not converged, increase N_{HF} (e.g., by 20%) and repeat.

C. LLM-based low-fidelity model

The low-fidelity model is trained on outputs generated by Large Language Models, hence the designation "low-fidelity" to reflect this derived data source. LLMs such as GPT-4 contain extensive implicit physics knowledge and empirical manufacturing relationships within their training data. This latent knowledge is leveraged to develop a low-fidelity estimator through carefully designed prompt engineering, as illustrated in the referenced figure. The architecture creates an effective framework for utilizing this embedded knowledge, including:

- *Knowledge Extraction*: Access implicit physical relationships between parameters
- *Low-Cost Estimation*: Generate predictions without physical experiments
- *Variance Control*: Ensure consistency through multi-round prompting

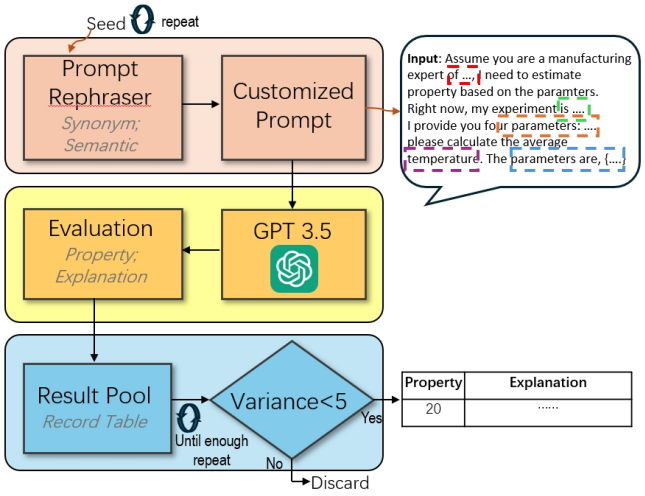


Fig. 3. LLM-based low-fidelity model workflow: (1) Initial prompt engineering, (2) Semantic rephrasing, (3) Variance-controlled response generation, (4) Integration with BO framework

A critical consideration for the low-fidelity model is crafting an effective prompt for the LLM, which should in-

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corporate situation context, parameter settings, and expected outcomes in a structured format, such as "As a manufacturing expert, estimate [property] given: Parameter 1: [value] ± [tolerance] ... Consider material properties, physical laws, and typical process relationships."

Despite the valuable insights provided by LLMs, their outputs may exhibit biases stemming from prompt formulation, necessitating our Rephraser Mechanism to refine and standardize responses for consistent optimization performance, as shown in Fig. 3. The Rephraser Mechanism generates N variants through:

$$P_i = \text{Rephrase}(P_{\text{base}}) \quad i = 1, ..., N \tag{17}$$

where P_i maintains semantic equivalence through similar replacement, like "estimate" $\rightarrow$ "calculate" or "Given X, find Y" $\rightarrow$ "Determine Y when X is".

1) Variance-Controlled Estimation: For input parameters x, we collect M responses $y_{LLM}^{(1)}, ..., y_{LLM}^{(M)}$ through multi-round querying:

$$\bar{y}\text{LLM} = \frac{1}{M} \sum i = 1^M y_{\text{LLM}}^{(i)}, \tag{18}$$

$$\sigma_{\text{LLM}}^2 = \frac{1}{M} \sum_{i=1}^{M} (y_{\text{LLM}}^{(i)} - \bar{y}_{\text{LLM}})^2 \tag{19}$$

Acceptance Criteria :

$$\sigma_{\text{LLM}}^2 < \tau \quad (\tau = 5 \text{ in implementation}) \tag{20}$$

2) Integration with Bayesian Optimization: The LLM estimates form our initial low-fidelity dataset:

$$\mathcal{D}\text{LLM} = (\mathbf{x}i, \bar{y}\text{LLM}, i)i = 1^n \tag{21}$$

These are subsequently refined by the residual GP:

$$\delta(\mathbf{x}) = f_{\text{high}}(\mathbf{x}) - f_{\text{LLM}}(\mathbf{x}) \sim \mathcal{GP}(0, k_{\text{high}}) \tag{22}$$

V. Case Study

Aircraft cabin assembly increasingly relies on human-robot collaboration (HRC) to address critical challenges in precision drilling operations. Large industrial robots, while essential for high-risk tasks like structural drilling, often lack sufficient historical data in dynamic assembly environments, leading to inefficiencies (e.g., frequent recalibration) and safety-critical errors during human-robot interactions. This necessitates HRC to synergize robotic precision (±0.1mm accuracy) with human contextual adaptability, particularly for drilling composite materials with unpredictable variations.

Precision parameter optimization for robotic drilling arms is critical to ensure hole quality and prevent composite delamination in aircraft structures. However, data scarcity in HRC environments—stemming from variable material properties and dynamic human interventions—severely limits traditional optimization methods requiring extensive high-fidelity experiments.

A. Experiment setting

A high-fidelity drilling process model is established to evaluate critical parameters influencing hole quality in aircraft cabin assembly. The physics-based simulation integrates five key operational parameters: Maximum drilling force, Spindle rotational speed, System vibration sensitivity, Surface interaction coefficient, Thermal conductivity. The model outputs surface roughness (μm) for the hole quality evaluation.

We developed a finite element method (FEM)-based high-fidelity drilling simulator incorporating critical physical phenomena [22] [23]: tool-workpiece contact dynamics, thermal-mechanical coupling, and composite material delamination thresholds. The model captures nonlinear interactions between drilling force, spindle kinematics, and material properties using empirical relationships from aerospace machining research, providing a physics-grounded evaluation of surface roughness under varying process parameters.

For the low-fidelity model, we leverage outputs generated by LLMs to train the BO. This innovative approach addresses the data scarcity challenge by utilizing the LLM's ability to reason about physical processes based on embedded knowledge, while maintaining computational efficiency.

B. Experiment result

Fig. 4 demonstrates the convergence of a GP optimization model in minimizing surface roughness (μm) during drill pressure tuning, plotting iterations against surface roughness. Initially, high roughness reflects suboptimal parameters, but by iter 100, the GP predictions closely align with the "True" values, indicating successful convergence.

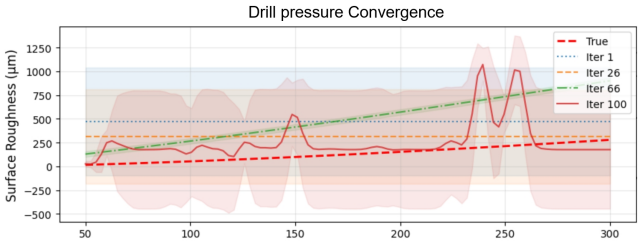
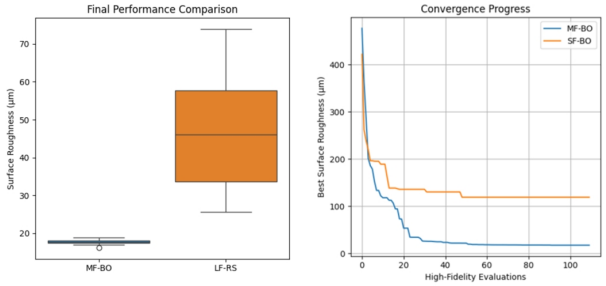


Fig. 4. drill pressure convergence

In order to evaluated the advanced of the proposed method, purely low-fidelity strategies, Low-Fidelity Random Search (LF-RS) is utlized to compare with proposed method, Multi-Fidelity Bayesian Optimization (MF-BO). This comparison aims to quantify the trade-offs between computational efficiency, optimization accuracy, and practical applicability across these methodologies, particularly in precision-sensitive domains like machining processes, where minimizing surface roughness is critical for functional performance.

As illustrated in Figure Fig. 5 (a), MF-BO achieves the lowest surface roughness, outperforming LF-RS. The narrow interquartile range of MF-BO indicates robust convergence, whereasLF-RS exhibits higher variability due to constrained exploration under limited high-fidelity evaluations. LF-RS

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(a) Performance comparison (b) Convergence process

Fig. 5. Performance and convergence comparison

demonstrates the poorest performance, underscoring its susceptibility to random sampling inefficiencies. These results highlight MF-BO's ability to synergize multi-fidelity data for resource-efficient optimization, making it particularly advantageous for precision-critical applications such as aerospace component manufacturing, where minimizing surface roughness directly enhances functional longevity and operational reliability.

Fig. 5 (b) demonstrates that the proposed MF-BO achieves superior performance compared to LF-RS in minimizing surface roughness with fewer high-fidelity evaluations. MF-BO strategically integrates low-fidelity data to approximate high-fidelity trends, enabling efficient exploration of the design space and prioritizing promising regions for costly high-fidelity validation. In contrast, LF-RS relies on random sampling of low-fidelity data, lacking systematic guidance, which results in slower convergence and higher resource consumption for equivalent performance.

VI. CONCLUSIONS

This paper presents an LLM-Augmented Multi-Fidelity Bayesian Optimization framework that successfully addresses the parameter optimization challenges in HRCA. The aircraft cabin drilling case study demonstrates that our approach significantly reduces the required number of high-fidelity evaluations while maintaining high-quality optimization outcomes.

ACKNOWLEDGMENT

This research is partially funded by the General Research Fund (GRF) from the Research Grants Council of the Hong Kong Special Administrative Region, China (Project No.PolyU15210222 and PolyU15206723).

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